A Master's Course on Principles of Statistical Learning (2025)

This advanced course in statistical learning dives into modern methods and their theoretical foundations. While the core focus is on conceptual understanding and statistical principles, the material will be reinforced through empirical demonstrations using data.

This document provides a detailed course plan, including information on the main texts, tentative schedule, assessment structure, and recommended supplementary resources.

1 Prerequisites

Students are expected to have a strong[1] background in linear algebra, calculus, probability and statistics, and proficiency in a programming language suitable for scientific computing. A prior introductory course in machine learning or statistical modeling is highly recommended.

2 Course Materials

The main text for this course is:

- ***The Elements of Statistical Learning: Data Mining, Inference, and Prediction*** (ESL), by Trevor Hastie, Robert Tibshirani, and Jerome Friedman (2nd edition, corrected 12th printing, 2017).
  Available at: `https://hastie.su.domains/ElemStatLearn/`

ESL is widely regarded as the gold standard reference for statistical learning, offering comprehensive coverage of both theory and methodology. We aim to cover approximately 70% of the materials in ESL, supplemented with:

- ***An Introduction to Statistical Learning: With Applications in Python*** (ISL), by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani (2023).
  Available at: `https://www.statlearning.com/`
- ***Learning Theory From First Principles*** (Bach), by Francis Bach (final version, 2025).
  Available at: `https://www.di.ens.fr/~fbach/ltfp_book.pdf`

ISL emphasizes a conceptual and application-oriented understanding of statistical learning methods, while Bach provides a rigorous treatment of the underlying mathematical principles. Additional materials may also be used to supplement the main text.

Lecture slides and the associated code will be made available after each class.

3 Tentative Schedule

We have 14 lecture sessions and 7 tutorial sessions.

Date	Session	Topic	References

[1] At a minimum, you should be familiar with most of the material in Part I of: `https://mml-book.github.io/`.

Aug 26	**L1**	Introduction to Supervised Learning	ESL (Ch. 1–2), ISL (Ch. 1–2), Bach (Ch. 2)
Aug 28	**L2**	Linear Methods for Regression	ESL (Ch. 3), ISL (Ch. 3, Ch. 6), Bach (Ch. 3)
Aug 29*	**T1**	Tutorial	-
Sep 2	**L3**	Linear Methods for Classification	ESL (Ch. 4), ISL (Ch. 4)
Sep 4	**L4**	Maximal Margin Methods for Classification	ESL (Ch. 4, Ch. 12.1-12.3), ISL (Ch. 9), Bach (Ch. 4.1)
Sep 5	**T2**	Tutorial	-
Sep 9	**L5**	Beyond Linearity: Basis Expansions, Local Averaging, Kernel Smoothing	ESL (Ch. 5.1-5.7, Ch. 6), ISL (Ch. 7), Bach (Ch. 6)
Sep 11	**L6**	Kernel Methods: Fundamentals	Bach (Ch. 7), ESL (Ch. 5.8)
Sep 12	**T3**	Tutorial, **Project Proposal Due**	-
Sep 16	**L7**	Kernel Methods: Special Topics	Bach (Ch. 7), additional materials
Sep 18	**L8**	Model Assessment and Selection, Generalization Theory (Optional)	ESL (Ch. 7, 9.1), ISL (Ch. 5), Bach (Ch. 4)
Sep 19*	**T4**	Tutorial	-
Sep 23	**L9**	Tree-Based Methods and Ensembling Methods	ESL (Ch. 9.2-10, Ch. 15), ISL (Ch. 8), Bach (Ch. 10)
Sep 25	**L10**	Neural Networks: Fundamentals	ISL (Ch. 10), ESL (Ch. 11), Bach (Ch. 9)
Sep 26	**15**	Tutorial	-
Sep 30	**L11**	Rethinking Generalization: Modern Phenomena in Overparameterized Models	ISL (Ch. 10), ESL (Ch. 11), Bach (Ch. 12)
Oct 2	**L12**	Rethinking Reliability: Probabilistic Approaches for Reliable Models	ESL (Ch. 8, 11), Bach (Ch. 14), additional materials
Oct 3	**T6**	Tutorial	-
Oct 7	**L13**	Unsupervised Learning: Clustering and Mixture Models	ISL (Ch. 12), PRML (Ch. 9), ESL (Ch. 14)
Oct 9	**L14**	Unsupervised Learning: Dimensionality Reduction and Advanced Topics	ISL (Ch. 12), PRML (Ch. 12), ESL (Ch. 14), extra materials
Oct 10	**T7**	Tutorial	-
Oct 19	**-**	**Project Report Due**	-
Oct 23	**-**	**Exam**	-

4 Assessments and Grading

See Canvas.

5 Additional Resources

The following references may also be useful for deepening your understanding of the topics covered:

Machine Learning & Deep Learning:

- *Pattern Recognition and Machine Learning* (PRML), by C. Bishop (2006)
- *Deep Learning: Foundations and Concepts*, by C. Bishop and H. Bishop (2024)
- *Deep Learning*, by I. Goodfellow, Y. Bengio, A. Courville (2016)
- *Probabilistic Machine Learning: An Introduction*, by K. Murphy (2022)
- *Patterns, Predictions, and Actions*, by M. Hardt and B. Recht (2022)
- *Understanding Machine Learning: From Theory to Algorithms*, by S. Shalev-Shwartz and S. Ben-David (2014)
- *Foundations of Machine Learning*, by M. Mohri, A. Rostamizadeh, A. Talwalkar (2018)
- *Learning from Data*, by Y. Abu-Mostafa, M. Magdon-Ismail, H.T. Lin (2012)
- *Machine Learning: A First Course for Engineers and Scientists*, by A. Lindholm, N. Wahlström, F. Lindsten, T. Schön (2022)

Mathematical & Statistical Foundations:

- *Mathematics for Machine Learning*, by M. Deisenroth, A. Faisal, C. Ong (2020)
- *Convex Optimization*, by S. Boyd and L. Vandenberghe (2004)
- *Statistical Inference*, by G. Casella and R. Berger (2nd edition, 2002)
- *All of Statistics: A Concise Course in Statistical Inference*, by L. Wasserman (2004)
- *All of Nonparametric Statistics*, by L. Wasserman (2006)
- *Large-Scale Inference: Empirical Bayes Methods for Estimation, Testing, and Prediction*, by B. Efron (2010)
- *Bayesian Data Analysis*, by A. Gelman, J. Carlin, H. Stern, D. Dunson, A. Vehtari, D. Rubin (3rd edition, 2013)
- *Information Theory, Inference, and Learning Algorithms*, by D. MacKay (2003)

Practical Implementation Resources:

- *Scikit-learn: Machine Learning in Python** (available at `https://scikit-learn.org/stable/`)
- *Dive into Deep Learning (D2L)**, by A. Zhang et al. (available at `https://d2l.ai/`)
- *Practical Deep Learning for Coders**, by J. Howard (at `https://course.fast.ai/`)
- *PyTorch in One Hour: From Tensors to Training Neural Networks on Multiple GPUs**, by S. Raschka (available at `https://sebastianraschka.com/teaching/pytorch-1h/`)